

Optimal Pressure Regulation in Water Distribution Systems Based on an Extended Model for Pressure Reducing Valves

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Abstract Optimal pressure regulation to reduce water losses in water distribution systems (WDSs) becomes an important concern due to the increasing water demand and the threat of drought in many areas of the world. The leakage amount in a WDS depends heavily on its operating pressure and thus can be minimized by implementing optimal pressure strategies through pressure reducing valves (PRVs). To achieve this, a model-based optimization is necessary, where an accurate model of the PRVs is required. The PRV models having been used until now for pressure regulations are two-mode models which cannot circumscribe many situations occurring in WDSs. In this paper, we extend the existing model by a three-mode one for PRVs which is able to describe the required circumstances of pressure regulations in WDSs. The non-smoothness of this model is smoothed by an approximation approach, thus allowing the formulation and solution of a continuous nonlinear optimization problem for optimal pressure regulation. Two benchmark WDSs are used to verify our approach and it can be shown from the results that our PRV model outperforms the existing models in terms of the quality and accuracy of the optimal solutions.

Keywords Pressure management · Leakage control · Water distribution system · Pressure reducing valve · Modeling · Optimization

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1 Introduction

Climate change, high temperature and the increase of domestic and industrial demand bring droughts, shortages and water restrictions to a considerable number of locations in the world. Moreover, the problem of excessive water losses in water distribution systems is common in many countries and it reaches a significant part of total water production (Karadirek et al. 2012). As a result, reduction and control of water losses has become a high priority to water utilities and regulators (Öztürk et al. 2007). Water losses in a WDS comprise real losses and apparent losses. Real losses are mainly caused by leakages at network fittings, pipe joints, breaks and/or bursts in pipes, whereas apparent losses are mainly due to illegal water consumption and inaccurate metering (Lambert 2002; Tabesh et al. 2009). Excessive water losses can lead to service interruption, energy losses and may cause contaminant intrusion into a pipe under low or even negative pressure conditions (Puust et al. 2010; Alkassseh et al. 2013).

Water leakages can be considered as additional demands at nodes and mathematically modeled as a proportional relation to the nodal pressure (Germanopoulos and Jowitt 1989). The leakage amount in a WDS increases significantly when operating at an excessive pressure (Savić and Walters 1996). For this reason, reducing the excessive pressure will lead to a reduction of the leakage amount and the risk of further leaks in a WDS (Ulanicki et al. 2000). To efficiently reduce water leakage in a WDS, the hydraulic model of a water distribution system with leakage terms should be calibrated. Tabesh et al. (2009) proposed to use an integrated model to evaluate losses in WDSs. A method was introduced to determine the nodal and pipe leakage by using a hydraulic model in EPANET. Karadirek et al. (2012) implemented pressure management to reduce water losses in a water distribution network in Antalya, Turkey using minimum night flows (MNF) combined with a hydraulic model in EPANET and optimized pressure settings of PRVs. Alkassseh et al. (2013) applied a statistical modeling approach to estimate the water losses in a city in Malaysia using MNF in which many factors affecting the water losses were considered. After having well calibrated the hydraulic model for a WDS, pressure management aiming at reducing water losses can be achieved by a model-based optimization. Nazif et al. (2010) optimized tank levels to reduce water losses in urban WDSs. An artificial neural network (ANN) model was developed to calculate the nodal pressures based on different water levels in tanks. The ANN model was then coupled with a genetic algorithm (GA) to minimize water leakage in an urban water distribution network.

Approaches to optimal localization and regulation of pressure reducing valves (PRVs) and/or isolated valves in WDSs to reduce excessive pressure have been recently investigated (Savić and Walters 1996; Vairavamorthy and Lumbers 1998; Ulanicki et al. 2008). The first step is to choose optimal positions for such valves to be installed in the WDS under consideration. Genetic algorithms, scatter search, and a MINLP approach were employed to optimize the localization of isolated valves and PRVs to minimize leakage flows in WDSs (Araujo et al. 2006; Liberatore and Sechi 2009; Creaco et al. 2010; Dai and Li 2014).

When the valves have been installed in the WDS, the optimal operation problem is to be considered to find optimal operating strategies. For the isolated valves the optimization task is to determine their on/off operations (Savic and Walters 1995), while for the PRVs their pressure settings (openings) are to be determined by the optimization (Vairavamorthy and Lumbers 1998). Due to the fact that PRVs can maintain pressures at downstream nodes at the preset levels regardless how large the pressures at the upstream nodes are, PRVs are more beneficial for the purpose of optimal pressure regulation (Ulanicki et al. 2008).

The minimization of excessive pressure by optimal operations of PRVs and/or isolated valves can be achieved by formulating and solving a nonlinear programming (NLP) problem (Vairavamoorthy and Lumbers 1998) and/or a mixed-integer nonlinear programming (MINLP) problem (Savic and Walters 1995), respectively. In Sterling and Bargiela (1984), Germanopoulos and Jowitt (1989), and Jowitt and Xu (1990), the method of successive linear programming was used to solve the NLP problem where the objective function was defined as the sum of differences between each nodal pressure and its minimum allowable pressure. It was an iterative approach that involves a linearization of the objective function and the constraints around the current solution and then a linear programming problem is solved to obtain the new solution. The procedure is repeated until a convergence criterion is satisfied.

Savic and Walters (1995) optimized the on/off operations of isolated valves to minimize the excessive pressure in a WDS using an evolutionary algorithm. Genetic algorithms combined with EPANET 2 (Rossman 2000) were also used to determine optimal outlet pressures of PRVs for optimal pressure regulations in WDSs (Awad et al. 2009; Mahdavi et al. 2010; AbdelMeguid and Ulanicki 2011). From the optimal solution, the outlet pressures of a PRV can be adjusted continuously by flow modulating schedule.

Vairavamoorthy and Lumbers (1998) formulated a NLP problem for optimal pressure regulation to minimize the leakage flow in WDSs, in which the nodal pressures are allowed to be lower than their minimum values by a minor violation in order to achieve a higher decrease of leakage amount in a WDS. Two different objective functions were used: 1) the sum of excessive pressures and 2) the sum of available leakages. It was shown that the use of the first option will lead to a higher reduction of leakage amount as compared with the use of the second option. Ulanicki et al. (2008) presented an approach to optimize the schedules of both boundary and internal PRVs for 24 hours to minimize the leakage flows in the Domestic Meter Areas in UK. As a result, the control flow modulation curves for the PRVs, i.e., the relations between the flows and the pressure settings which are essential for online PRV control, were deduced. In the above mentioned optimization approaches for optimal pressure regulation, the accuracy of the model used to describe the WDS under consideration plays an essential role. The head-flow relation for describing the pipeline and the PRV hydraulics represents the most important part in such a model. The model of a PRV located in a pipe has been commonly described by the Hazen-Williams equation (Vairavamoorthy and Lumbers 1998; Ulanicki et al. 2008), leading to a two-mode model. A variable denoting the valve setting, varying between zero (fully closed) and one (fully open) is usually introduced to represent the operation of a PRV. This model can only represent the *normal mode* when a PRV maintains the downstream pressure at the preset value and the *open mode* when the downstream pressure is lower than the pressure setting. However, it cannot account for the *check valve mode* for preventing reverse flows when upstream pressure is lower than the downstream pressure. The check valve mode is important for situations with varying water demand, i.e., the consumption of water during the night drops considerably (Sivakumar and Prasad 2015), which take place in most WDSs.

Recently, the authors of this paper studied the localization of pressure reducing valves (PRVs) in WDSs and developed an approach for solving the mixed-integer nonlinear programming (MINLP) problem (Dai and Li 2014). The locations of PRVs were to be optimally determined for a given WDS, i.e. a design problem was considered, where the model of PRVs for describing their operations was not required in that study. The current paper deals with the optimal operation problem of PRVs, after their positions have been determined. In this paper, we propose a non-smooth model, which is an extension of the existing

model, to describe the hydraulic behaviors of PRVs. This model is able to circumstantiate the three (open, normal, check valve) operational modes of a PRV. The non-smoothness of the model is smoothed by using an interior-point approximation approach, so that it can be employed for the formulation of a NLP problem for optimal pressure regulations of WDSs. Two benchmark WDSs are taken as case studies to demonstrate the advantages of using the extended PRV model. It can be shown from many scenarios of demand patterns that the optimal results by using our PRV model outperform those by using the previous models. In addition, a higher accuracy of the solution can be achieved with our PRV model in comparison to that by using the models from the literature.

The remainder of the paper is organized as follows. In Section 2, the shortcomings of the existing PRV models are analyzed and then an extended model is proposed. The formulation of a general optimization problem for optimal pressure regulation in WDSs is presented in Section 3. In Section 4, we apply the proposed PRV model to optimize pressure regulations for two benchmark WDSs. Conclusions of the paper are provided in Section 5.

2 Model of a Pressure Reducing Valve

2.1 The Existing Models

A pressure reducing valve is commonly described by the following equation (Vairavamoorthy and Lumbers 1998)

$$Q_{i,j} = R_{i,j} V_{i,j} \text{sign}(H_i - H_j) |H_i - H_j|^{0.54} \quad (1)$$

where

$$R_{i,j} = \frac{278.54 C_{i,j} D_{i,j}^{2.63}}{L_{i,j}^{0.54}} \quad (2)$$

$$0 \leq V_{i,j} \leq 1.0$$

where $R_{i,j}$ is the resistance of the valve; $C_{i,j}$, $D_{i,j}$, $L_{i,j}$ are the Hazen-Williams coefficient, the diameter, and the length of pipe ij ; H_i , H_j are nodal heads of node i and j , respectively; $V_{i,j}$ is the valve setting.

It can be seen that Eq. 1 is non-smooth and thus its gradient will be discontinuous. Therefore the optimization problem based on this model will also be non-smooth and cannot be solved directly by a NLP solver. For this reason, a smoothing strategy is needed. In addition, it is important to prevent the evaluation of the derivatives at the point of discontinuity, i.e., when the heads H_i and H_j at the two end nodes of the PRV are equal (Vairavamoorthy and Lumbers 1998).

Another form of a PRV model is described as (Ulanicki et al. 2008).

$$H_i - H_j = k_{i,j}(v_{i,j}) R_{i,j} Q_{i,j} |Q_{i,j}|^{0.852} \quad (3)$$

where $R_{i,j}$ is the resistance of PRV; $k_{i,j}(v_{i,j})$ is a resistance factor of the head loss related to valve opening $v_{i,j}$. $k_{i,j}(v_{i,j})$ can take a value from 1.0 to a very large positive number (i.e., infinity) (Ulanicki et al. 2008).

To use Eq. 3 in the formulation of the NLP problem, an approximation $Q_{i,j} \simeq (Q_{i,j}^2 + \tau^2)^{0.5}$ can be used where τ is a small value to smooth (3), so that it becomes

$$H_i - H_j = k_{i,j} (v_{i,j}) R_{i,j} Q_{i,j} (Q_{i,j}^2 + \tau^2)^{0.426} \quad (4)$$

It can be seen that Eq. 4 can be used only if PRV operates with condition $H_i \geq H_j$, i.e. in the normal or open modes. However, the diurnal demand pattern in a WDS can fluctuate arbitrarily (i.e., the water consumption decreases significantly at the night) and thus there may be the case where the PRVs are forced to shut off when $H_i < H_j$ (the check valve mode). Therefore, it is not suitable to use this model in the formulation of the optimization problem for optimal pressure regulation with varying demand patterns. From the numerical solution point of view, the NLP solver can fail to solve the optimization problem based on the two-mode model or converged to low quality solutions in many demand scenarios.

2.2 An Extended Model

In general, a PRV in a WDS can operate in one of the following three modes (Simpson 1999; Rossman 2000). First, it operates in a normal mode (mode 1), i.e., its resistance ($R_{i,j}$) is adjusted to maintain the preset pressure on its downstream side. Second, it operates as a fully opened valve (mode 2), when its downstream pressure cannot be maintained, i.e., the pressure on the upstream and downstream side is less than the pressure setting. Third, it acts as a check valve to prevent the water flow in a reverse direction (mode 3), i.e., when the pressure on the downstream side is higher than the one on the upstream side. Therefore, it is necessary to model a PRV which can describe all three modes. Considering a PRV on a pipe connecting node i with head H_i to node j with head H_j , we can use the following equation to describe its head loss when operating in mode 1 and 2 with flow $Q_{i,j}$ (Bhave and Gupta 2006; Piller and van Zyl 2014; Sivakumar and Prasad 2015)

$$H_i - H_j = R_{i,j} Q_{i,j}^2 \quad (5)$$

In the normal mode (mode 1), the resistance $R_{i,j}$ is adjusted to maintain the pressure at node j to a preset pressure setting. Mathematically, depending on the pressure setting, $R_{i,j}$ can range from a small value when the PRV acts as a fully opened valve to a very large value when it acts as a fully closed valve. In particular, in the open mode (mode 2), the resistance can be expressed as Bhave and Gupta (2006)

$$R_{i,j} = \frac{8K_{i,j}}{\pi^2 g D^4} \quad (6)$$

where $K_{i,j}$ is the head-loss coefficient of PRV and it can be taken as 10.0 (Fietz 1977; Jeppson and Davis 1977), D is the diameter of the PRV, g is the gravity of water. This means that in the open mode $R_{i,j}$ is fixed. In the normal mode (mode 1), we have

$$R_{i,j} > \frac{8K_{i,j}}{\pi^2 g D^4} \quad (7)$$

To represent the operation of PRV in the normal mode, we introduce a variable coefficient $v_{i,j}$ with $0 < v_{i,j} \leq 1$ to account for the variable resistance, so that Eq. 5 becomes

$$H_i - H_j = \frac{1}{v_{i,j}} R_{i,j} Q_{i,j}^2 \quad (8)$$

where $Q_{i,j} \geq 0$, i.e., the flow through the valve should be kept in one direction. The boundaries of this coefficient correspond to the two limiting operation cases of the normal mode of a PRV, namely it is fully closed in the case of $v_{i,j} \rightarrow 0$ and is fully opened in the case of $v_{i,j} = 1$.

Similar to Eq. 4, Eq. 8 can only describe mode 1 and mode 2 of PRVs. In the check valve mode (mode 3), the PRV should be fully closed and thus the flow though it is zero, i.e. $Q_{i,j} = 0$. To consider the operation mode 3 where a PRV acts to prevent a reverse flow when $H_i < H_j$, we extend Eq. 8 in the following form

$$\max(0, H_i - H_j) = \frac{1}{v_{i,j}} R_{i,j} Q_{i,j}^2 \quad (9)$$

This PRV model is able to describe all three operation modes required in optimal pressure regulation. However, Eq. 9 is a non-smooth equation and hence cannot be directly used for the formulation of a NLP problem. In this study, we propose to use the interior-point smoothing approach proposed by Gopal and Biegler (1999) to approximate the left-hand side of Eq. 9 by the following function

$$\max(0, H_i - H_j) \simeq \frac{(H_i - H_j) + \sqrt{(H_i - H_j)^2 + \tau^2}}{2} \quad (10)$$

where τ is a small value and chosen as 0.001. In this way, the smooth model which is able to describe the three operational modes of a PRV is expressed as

$$(H_i - H_j) + \sqrt{(H_i - H_j)^2 + \tau^2} = \frac{2}{v_{i,j}} R_{i,j} Q_{i,j}^2 \quad (11)$$

In the next section, we will integrate the models in the forms of Eqs. 4, 8, and 11 in the formulation of NLP problems for optimal pressure regulation in WDSs. Results from case studies in section 4 will show the differences of the three models in terms of accuracy and feasibility.

3 Problem Formulation for Optimal Pressure Regulation

The aim of our optimization is to find optimal pressure settings for PRVs so as to minimize the excessive pressure in a WDS. Therefore, the objective function is defined as the excessive pressure at all nodes in the WDS in the optimization time horizon (Germanopoulos and Jowitt 1989)

$$\min F = \sum_{i=1}^{NJ} \sum_{k=1}^T (H_{i,k} - H_{i,k}^L) \quad (12)$$

where NJ is the total number of junction nodes, $T=24$ hours is the time horizon, $k = 1, \dots, 24$ is time interval, $H_{i,k}$ and $H_{i,k}^L$ are nodal head of node i and its prespecified minimum allowable head, respectively.

We consider a WDS with NJ junction nodes, NP pipes, N_r reservoirs and N_{prv} pressure reducing valves. The equality constraints consist of the following equations.

The continuity equation at nodes i

$$\sum_{j,k} Q_{j,i,k} - d_{i,k} - l_{i,k} = 0; i = 1, \dots, NJ \quad (13)$$

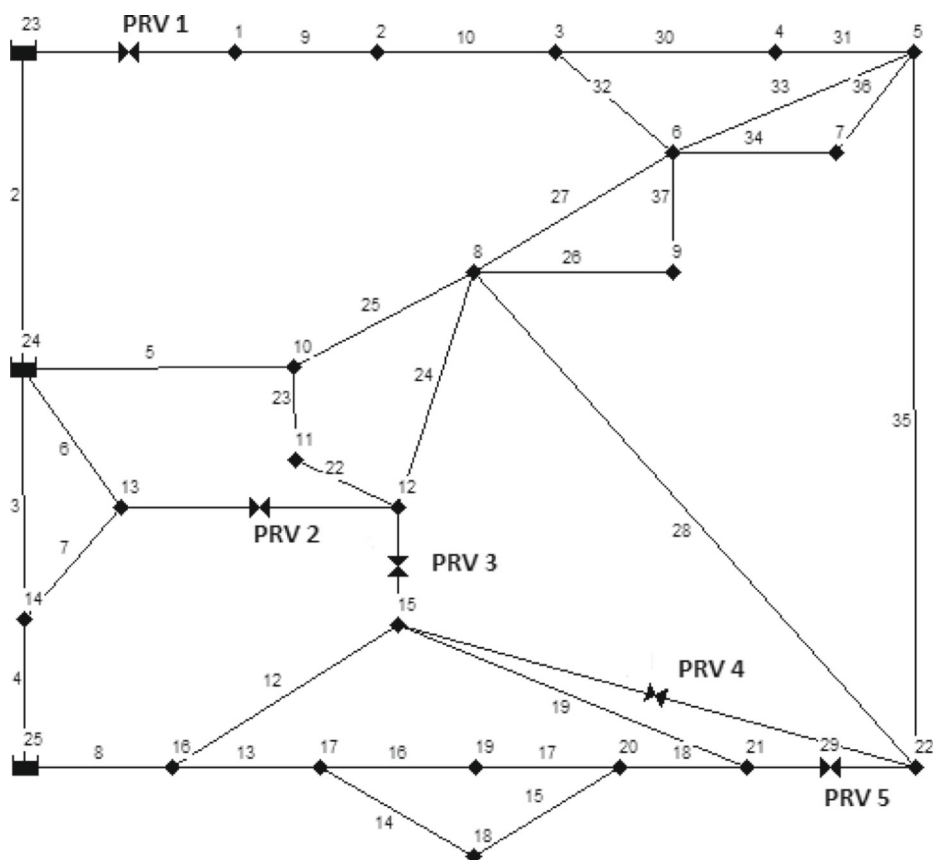


Fig. 1 A water distribution system with 5 PRVs (Sterling and Bargiela 1984)

where $d_{i,k}$ and $l_{i,k}$ are the demand and leakage amount associated to node i at time interval k , respectively; $Q_{j,i,k}$ is the flow through pipe ji . The energy equations for pipes connecting node i to node j

$$-H_{i,k} + H_{j,k} + \Delta H_{i,j,k} = 0; i, j = 1, \dots, NP \quad (14)$$

where the head loss $\Delta H_{i,j,k}$ through a pipe connecting node i to node j is computed either by the Hazen-Williams equation (Bhave and Gupta 2006)

$$\Delta H_{i,j,k} = \frac{10.67 L_{i,j}}{D_{i,j}^{4.87}} \left(\frac{Q_{i,j,k}}{C_{i,j}} \right)^{1.852} \quad (15)$$

or by the Darcy-Weisbach equation (Bhave and Gupta 2006)

$$\Delta H_{i,j,k} = \frac{8 L_{i,j} f}{g \pi^2 D_{i,j}^5} |Q_{i,j,k}| Q_{i,j,k} \quad (16)$$

The friction factor f in Eq. 16 is implicitly calculated using the Colebrook-White equation (Bhave and Gupta 2006)

$$\frac{1}{\sqrt{f}} = -2 \log \left(\frac{(\varepsilon/D)}{3.7} + \frac{2.51}{\text{Re} \sqrt{f}} \right) \quad (17)$$

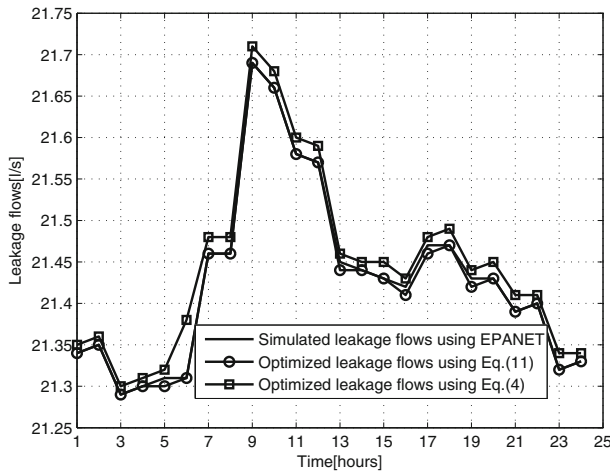


Fig. 2 Simulated and optimized leakage flows

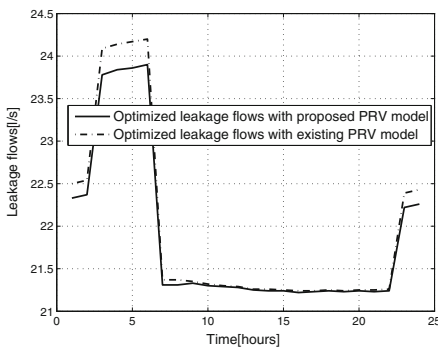
where Re is the Reynolds number

$$Re = \frac{VD}{\nu} \quad (18)$$

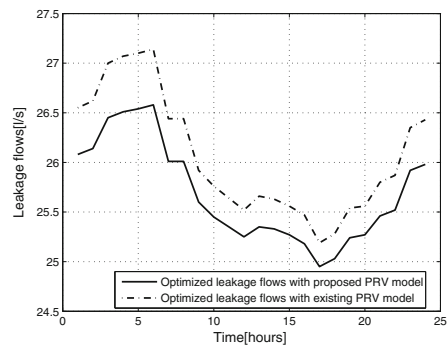
In Eqs. 15 and 16 L , D and C are the length, diameter and Hazen-William coefficient of pipe, respectively; ε in Eq. 17 is the pipe roughness coefficient; V and ν are the velocity of flow through the pipe and the viscosity of water, respectively.

The difference between Eqs. 15 and 16 lies in the fact that Eq. 15 is a simple and smooth model with a lower accuracy (Liou 1998), while the accuracy of Eq. 16 is much higher but it is more complicated and non-smooth. To use Eq. 16 in the formulation of a NLP, we use the smooth form proposed by Burgschweiger et al. (2009). The leakage amount $l_{i,k}$ associated to node i in Eq. 13 is calculated by Araujo et al. (2006).

$$l_{i,k} = C_L L_{t,i} p_{i,k}^\gamma \quad (19)$$



(a) demand factor = 0.5



(b) demand factor = 0.1

Fig. 3 Leak flows for two demand factors

where $L_{t,i}$ is the total length of pipes connected to node i , calculated by

$$L_{t,i} = 0.5 \sum_j L_{i,j} \quad (20)$$

In Eq. 19 C_L is the discharge coefficient of the orifice; γ is the pressure exponent and it may vary in a range of 0.5 to 2.5 (Lambert 1997; Tabesh et al. 2009); $L_{i,j}$ is the length of pipe ij ; $p_{i,k}$ is the pressure at node i .

The energy equations for PRVs located in the pipe from node i to node j (see Eq. 11)

$$(H_i - H_j) + \sqrt{(H_i - H_j)^2 + \tau^2} - \frac{2}{v_{i,j}} R_{i,j} Q_{i,j}^2 = 0 \quad (21)$$

where $R_{i,j}$ is calculated by Eq. 6.

The head of reservoir i is considered as a constant ($\bar{H}_i$), i.e.

$$H_{i,k} - \bar{H}_i = 0; i = 1, \dots, N_r \quad (22)$$

The inequality constraints consist of following operational restrictions.

Nodal head constraints

$$H^L \leq H_{j,k} \leq H^U; j = 1, \dots, NJ \quad (23)$$

Pipe flow constraints

$$Q^L \leq Q_{i,j,k} \leq Q^U; i, j = 1, \dots, NP \quad (24)$$

Valve flow constraints

$$0 \leq Q_{i,j,k} \leq Q^U; i, j = 1, \dots, N_{prv} \quad (25)$$

Valve variable coefficient constraints

$$0 < v_{i,j,k} \leq 1; i, j = 1, \dots, N_{prv} \quad (26)$$

In Eqs. 23–25 H^L , H^U , Q^L , and Q^U represent the lower and upper bounds of nodal heads and pipe flows, respectively. We use a well-known NLP solver based on the interior-point method, IPOPT (Wächter and Biegler 2006), to solve the nonlinear optimization problem formulated above. The Jacobian of the constraints as well as the gradient of the objective function are calculated and supplied to IPOPT. All the computation experiments in the following case studies are carried out on an Intel (R) Core (TM) 3.40GHz 2.99GB RAM desktop. For each node, a given base demand value for the time horizon is assumed and a pattern factor is defined for each hour for the demand profile (this is the commonly used way to define demand flows, e.g. in EPANET). Given these demands for all nodes and all hours as inputs of the optimization problem, our approach transfers it to a large-scale nonlinear programming (NLP) problem which needs to be solved only once. It means that the temporal and spatial changes of demands are considered simultaneously in our approach for minimizing the excessive pressure. Due to the high efficiency of NLP solvers (e.g., IPOPT) such problems can be solved in a short CPU time.

4 Case Studies

4.1 Case Study 1

At first, we apply our PRV model for optimal PRV pressure settings in a water distribution system as depicted in Fig. 1. With 37 links, 22 nodes and 3 reservoirs, this WDS was used as a case study for the PRV localization in Araujo et al. (2006). We use the Hazen-Williams

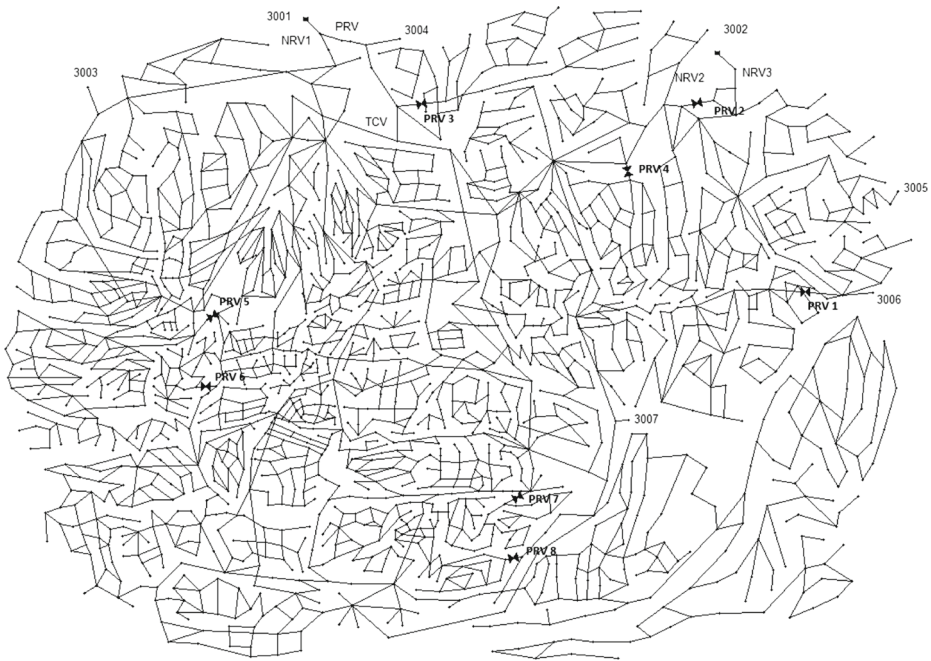


Fig. 4 EXNET water distribution system (Farmani et al. 2004)

equations for the head loss calculation. We consider 5 PRVs located in links 1, 11, 20, 21, and 29 as suggested in Dai and Li (2014), as shown in Fig. 1. The data for the demand pattern and reservoir heads for 24 hours, the discharge coefficient C_L and the leakage exponent parameter γ are from Araujo et al. (2006). The lower bounds for the pressure at the nodes are 30.0 (m). The optimal pressure regulation problem is formulated for 24 hours and the

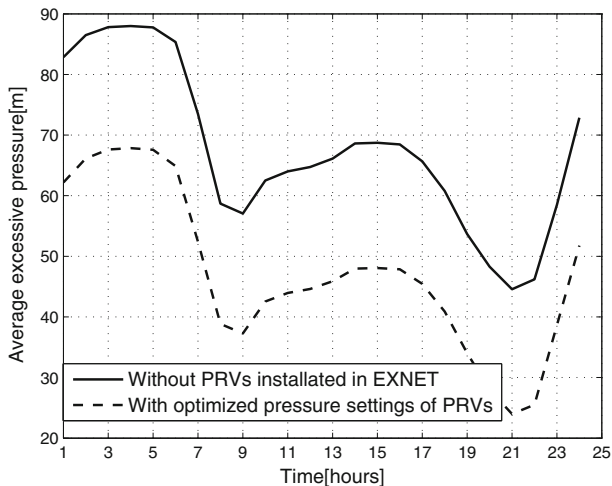


Fig. 5 Average excessive pressure with (dotted line) and without optimal pressure regulation (solid line)

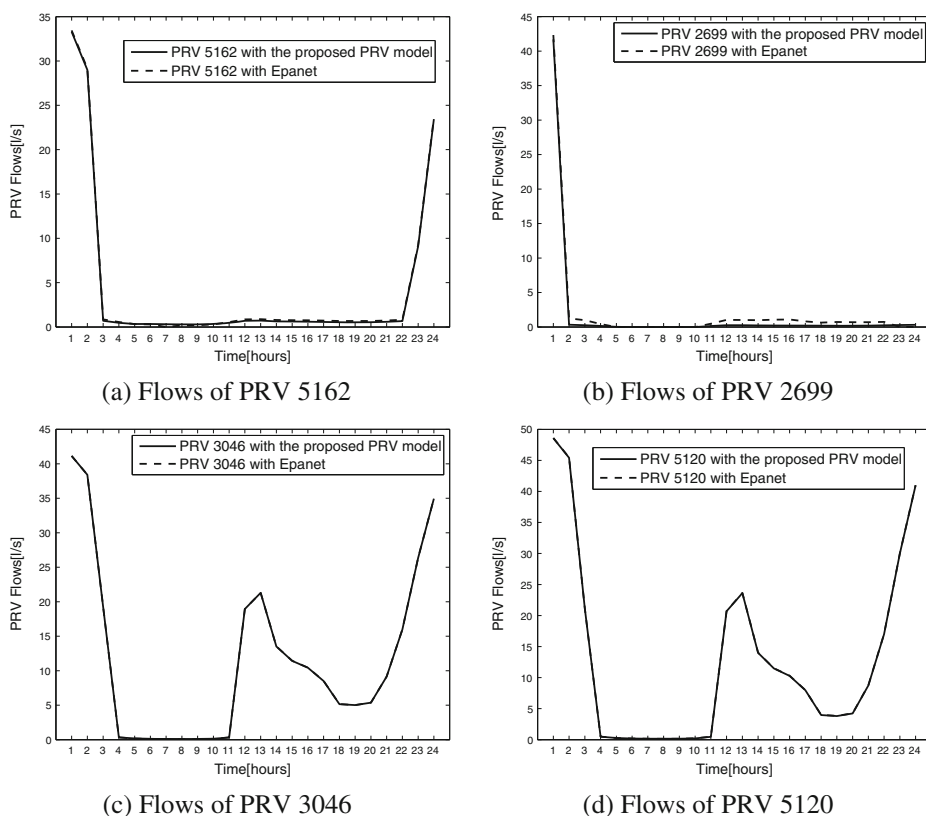


Fig. 6 Comparisons of PRV flows

resulting NLP has 1608 equality and inequality constraints and 1800 variables. For comparison, in addition to the diurnal demand pattern used in Araujo et al. (2006), we consider two additional demand patterns defined by multiplying 0.1 and 0.5 to the diurnal demand pattern, respectively. The idea of using different factors is to test the effect of the extended model with significant changes of demands. In general, any factor values can be chosen for the test. But the check valve mode (the extended operating mode in our approach) will be more relevant, if we consider lower demands. In addition, considering such varying demand factors allows to illustrate the advantages of the extended PRV model over the existing one in solving optimal pressure regulating problems.

IPOPT took an average of 4.53 (s) for solving the optimization problems. Corresponding to the diurnal demand pattern, the optimal solution reveals that the PRVs on links 1, 11, and 21 operate at appropriate pressure settings, while the PRVs on links 20 and 29 are fully closed in the whole 24 hours. As a result, the optimal operations of the PRVs lead to an average decrease of leakage amount for 5.72 (l/s) and an average decrease of the excessive pressure for 155.57(m), respectively, as compared with the case in which no pressure management (i.e., no PRV is installed in the WDS) is made.

The accuracy of our proposed PRV model is compared with the one in EPANET 2 by passing the optimal pressure settings for PRVs to EPANET 2 for simulation. It can be seen from Fig. 2 that the leakage flows from the EPANET model are almost the same as those

from our model. In addition, according to the simulation results, it is shown that the largest absolute discrepancy for the flows is less than 0.25(l/s) and it is about 0.022(m) for the heads. Therefore, it can be concluded that our extended PRV model is almost as accurate as the one used in EPANET 2.

To compare our PRV model with the existing model in the form of Eq. 4, we formulate the same optimization problem but with Eq. 4 as the PRV model in which $k_{i,j}(v_{i,j})$ is bounded by 1.0 and 1.E+20 (Ulanicki et al. 2008). Then, we solve both optimization problems by IPOPT. It can be seen in Fig. 2 that the existing model results in a bit higher leakage flows than those resulted from our model. This is due to the fact that, according to the optimization results, the flows through PRVs on links 29 and 20 are zero (i.e., they are closed) based on our PRV model, while they have small values (i.e., they operate in mode 1) based on the existing PRV model. In addition, as we pass the optimal PRV settings from the existing model to EPANET for simulation, EPANET 2 gives warnings that the system is unbalanced, while the solution from our PRV model is feasible for EPANET 2.

In the case of demand factor with 0.1, the optimization based on our extended PRV model (11) results in the objective function value as 5822.821 (m), while it is 6079.166 (m) based on the existing PRV model (4). In the case of demand factor with 0.5, our approach results in the objective function value as 2951.945 (m), while it is 3026.338 (m) using the existing PRV model. It can be seen that for both demand patterns our PRV model leads to a lower objective function value (i.e., lower excessive pressure) than that by the existing PRV model. This is because the optimization problem based on Eq. 4 has a smaller feasible region, i.e., only when $k_{i,j}$ has a very large value (except for the case $H_i = H_j$) there will be a feasible solution for a PRV with $Q_{i,j} = 0$. On the contrary, using Eq. 11, a feasible solution for a PRV can be $Q_{i,j} = 0$ without requiring a very small value of $v_{i,j}$ (except the case $H_i = H_j$). The leak flows found by solving optimization problems corresponding to demand factors of 0.5 and 0.1 are shown in Fig. 3. Again, it can be seen for both cases that our PRV model leads to higher decreases of leakage flows.

4.2 Case Study 2

Now we apply our PRV model to minimize the excessive pressure in the EXNET network with 1892 nodes and 2465 links as depicted in Fig. 4 (Farmani et al. 2004). To carry out optimal pressure regulation for this network, several modifications of the original model from Farmani et al. (2004) are necessary. The existing TCV and PRV in the network are replaced with pipes (diameter: 1000 mm for TCV and 500 mm for PRV, length: both with 10 m, roughness coefficient: 1.5 (mm) for TCV and 0.1 mm for PRV). The heads of reservoirs at nodes 3001 and 3002 are set to 80.0 (m) to avoid negative nodal pressures (Eck and Mevissen 2012). To regulate the excessive pressures, we consider 8 PRVs placed on links 5162, 2699, 3046, 5120, 3783, 2700, 4186, and 2938 as suggested in Dai and Li (2014), as shown in Fig. 4.

In the EXNET network, at nodes 3003, 3004, 3005, 3006, and 3007 water is supplied from adjacent systems with the average flow rate values of 63.00, 1388.0, 10.77970, 926.0001, and 26.1027 (l/s), respectively (Farmani et al. 2004). We assume that the demand at these nodes is kept constant with these values for 24 hours. For all other nodes, we use the same base demand data and with the daily water pattern for 24 h taken from Farmani et al. (2004). The lower bounds for pressures at all demand nodes are set to be 8.0 (m) (Eck and Mevissen 2012). Since the pipe hydraulics in EXNET is modelled by the non-smooth Darcy-Weisbach equation, we use the method by Burgschweiger et al. (2009) to smooth the model, thus allowing formulating a continuous optimization problem. The resulting NLP

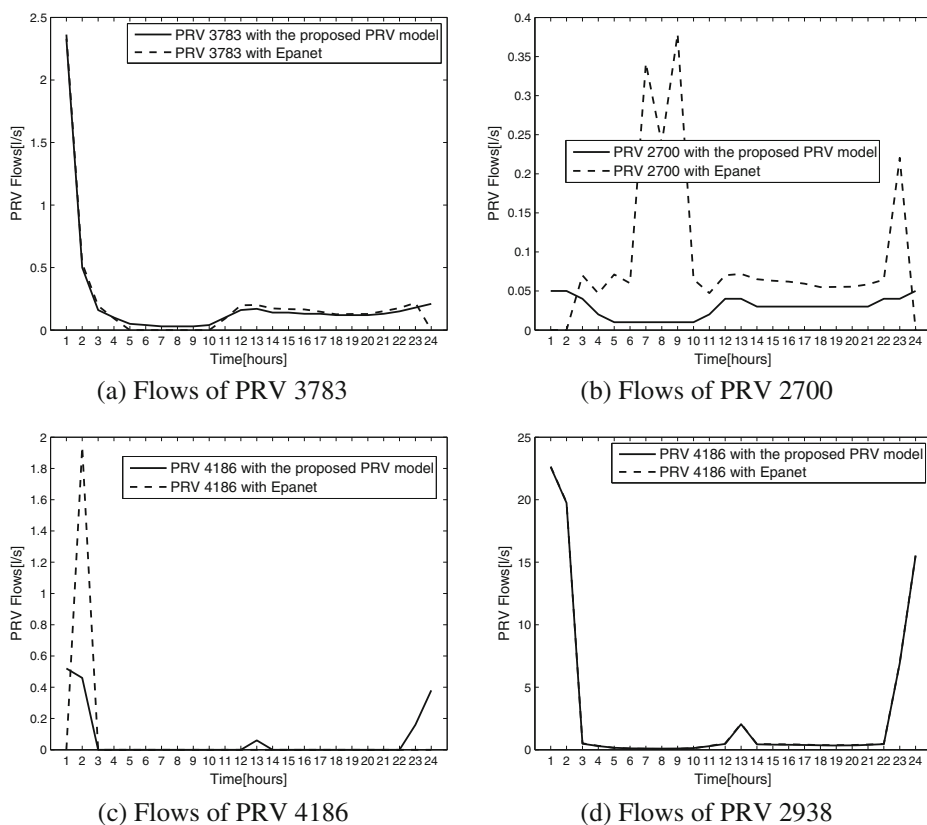


Fig. 7 Comparisons of PRV flows

has totally 104520 inequality and equality constraints and 104760 variables, respectively. It took only 237.195 s to solve this huge NLP problem.

The average excessive pressures for 24 hours both with (dotted line) and without optimal pressure regulation (solid line) are computed using EPANET which are shown in Fig. 5. The solid line is generated by the simulation of the network with the same demand patterns, but without considering PRVs. It can be seen that with the optimal pressure regulation by controlling PRV operations, a significant reduction of the average excessive pressure is achieved. This pressure reduction not only leads to a significant decrease of the existing leakage flow rates, but also limits the risk of further leaks and incidences of pipe bursts.

It can be also seen from Fig. 5 that the average excessive pressure will decrease when the demand increases (e.g., at time 21:00, demand factor = 1.0). This is because, when more water is taken out at the nodes, the nodal pressures will be reduced and thus the excessive pressures ($H_i - H^L$) decreased. Reversely, when the demand is small during the night, the excessive pressure will be high and thus the leakage amount will be increased. In addition, the pressure settings for the PRVs with a lower demand pattern factor tend to be higher than those with a higher demand pattern factor.

The optimal flows through the PRVs by the proposed (solid line) and the EPANET (dotted line) model are given in Figs. 6 and 7. It can be seen that the discrepancies of the flows are quite small. In particular, the flows through the PRVs on the links 5162, 2699, 3046,

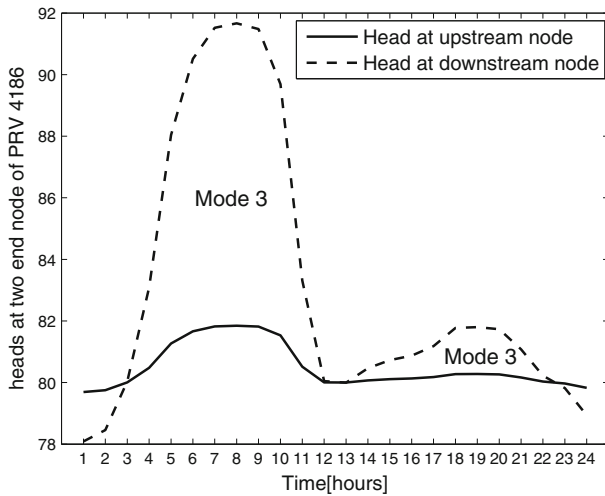


Fig. 8 Heads at upstream and downstream nodes of PRV 4186

5120, 3783, and 2938 by our model are almost the same as the ones by EPANET 2. The discrepancies in flows of the PRVs on links 2700 and 4186 in several time intervals, as shown in Fig. 7b and c, do not affect much to the overall operation of the network since their values are very small (i.e., 1.8 and 0.37 l/s). Moreover, through a comparison of pipe flows and nodal heads resulted from our optimization model and those from EPANET 2, the largest absolute discrepancy of nodal heads is smaller than 0.75 (m), and the largest relative discrepancy is 5.7 % and most of them is less than 2 %. Similarly, for flows larger than 10 (l/s), the largest relative discrepancy is 5 % and the most of them is less than 2.5 %. Therefore, it can be concluded that the hydraulic model used in our optimization is sufficiently accurate.

To demonstrate the advantage of our PRV model over the existing one, we also formulate an optimization problem based on Eq. 8. However, the NLP solver fails to solve the optimization problem in which 24 hours are considered. It can only solve the single optimization problem for several individual demand factors such as 1.0, 0.971, 0.933, 0.830, and 0.763 (see the demand pattern in Farmani et al. (2004)). For demand factors smaller than 0.763, it fails, since the condition $H_i - H_j \geq 0$ for the PRVs using Eq. 8 cannot be satisfied, i.e., the model cannot describe operation mode 3 (the check valve mode). To see the operation of the PRVs using our model, we now take the PRV on the link 4186 as an example. As shown in Fig. 8, the head (H_i) at the upstream node is lower than that at the downstream node (H_j) from 3:00 to 11:00 and from 13:00 to 22:00. It means that during these time periods, this PRV operates in mode 3 that cannot be handled by Eq. 8.

5 Conclusion

The task for the minimization of water leakage in WDSs poses a significant challenge. In this paper, we present a systematic approach for optimal pressure regulation in WDSs by optimizing control of PRVs. An extended model for properly describing the three-mode behaviors of PRVs is proposed. This non-smooth model is then smoothed by using an

interior-point approximation method, so that a continuous optimization problem is formulated and solved by a NLP solver. Based on the proposed PRV model, the minimization of the excessive pressures is carried out for two benchmark WDSs and the optimal results are compared with those obtained by using the existing PRV models. It can be shown that our PRV model is more beneficial and robust than the existing ones in terms of both quality and accuracy of the optimal solutions for many demand scenarios.

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